Angular velocity $\omega = 5 \text{ rad s}^{-1}$

Solution:

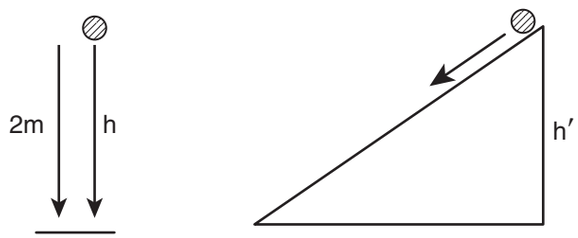
Angular momentum $L = I\omega = \frac{2}{5}mr^2\omega$

$$= \frac{2}{5} \times 20 \times (0.25)^2 \times 5 = 40 \times (0.0625) = 2.5$$

$$L = 2.5 \text{ kg m}^2\text{s}^{-1}$$

4. A solid cylinder when dropped from a height of 2 m acquires a velocity while reaching the ground. If the same cylinder is rolled down from the top of an inclined plane to reach the ground with same velocity, what must be the height of the inclined plane? Also compute the velocity.

Solution:



In the first case,

potential energy = kinetic energy

$$mgh = \frac{1}{2}mv^2$$

$$mg \times 2 = \frac{1}{2}mv^2 \qquad (1)$$

In second case,

potential energy = translational kinetic energy + rotational kinetic energy

$$mgh' = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$mgh' = \frac{1}{2}mv^2 + \frac{1}{2}\left(\frac{mr^2}{2}\right)\left(\frac{v^2}{r^2}\right)$$

$$\therefore mgh' = \frac{3}{4}mv^2 \qquad (2)$$

Dividing (2) by (1),

$$\frac{mgh'}{mg \times 2} = \frac{\frac{3}{4}mv^2}{\frac{1}{2}mv^2} = \frac{3}{4} \times \frac{2}{1} = \frac{3}{2}$$

$$h' = 3\,m$$

From equation (1), $2mg = \frac{1}{2}mv^2$

$$v = \sqrt{4g} = 2\sqrt{g}$$

$$v = 2 \times \sqrt{9.81}$$

$$v = 6.3\,m\,s^{-1}$$

5. A small particle of mass m is projected with an initial velocity v at an angle θ with x axis in X-Y plane as shown in Figure. Find the angular momentum of the particle.

Appendix-1.indd 285 06-01-2022 17:55:38